

Ajay Kumar Jha

Curriculum Vitae

QBB 258 A23, Department of Computer Science
North Dakota State University
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🌐 <https://hifromajay.github.io/>

Research Interests

Software Engineering, Software Testing and Maintenance, Mining Software Repositories, and Empirical Software Engineering.

Professional Experience

Assistant Professor (Aug 2022 – Present)

Department of Computer Science, North Dakota State University, Fargo, ND, USA

Postdoctoral Researcher (Mar 2020 – Jul 2022)

University of Alberta, Edmonton, Alberta, Canada (Advisor: Dr. Sarah Nadi)

Postdoctoral Researcher (Mar 2017 – Feb 2020)

Kyungpook National University, Daegu, Republic of Korea (Advisor: Dr. Woo Jin Lee)

Co-founder (Feb 2009 – Jun 2011)

Unified Technology Consultancy, Kathmandu, Nepal

Account Manager (Mar 2008 – Aug 2008)

V2Soft, Bangalore, India

Business Development Manager (Sep 2007 – Jan 2008)

FoxxsysTech, Gurgaon, India

Co-founder (Nov 2005 – Aug 2007)

MindproSoft, Bangalore, India

Education

Ph.D. in Computer Science and Engineering, Feb 2017

Kyungpook National University, Republic of Korea (Advisor: Dr. Woo Jin Lee)

M.S. in Computer Science, Aug 2013

Kyungpook National University, Republic of Korea (Advisor: Dr. Woo Jin Lee)

B.Sc. in Information Technology, Jan 2004

Sikkim Manipal University of Health, Medical and Technological Sciences, India

Research Funding

NSF CAREER Award (Jan 2027 – Dec 2031), PI, \$500,000

CAREER: Improving Software Test Quality and Developer Skills through Developer-in-the-Loop Test Optimization

USDA NIFA (May 2026 – May 2029), Co-PI, \$591,500

A-SYSRV: Systems Biology Reverse Vaccinology, An Integrated Vaccine Prediction Tool for Aquaculture Pathogens

USAID (Cancelled due to USAID shutdown), Co-PI, \$481,225

Digital Tools for Optimal Fish Feed Formulation in Nigeria

ND EPSCoR (Oct 2022 – Jun 2024), PI, \$20,000

Unit Test Recommendation and Reuse

Publications (*My Student at NDSU, **Co-advised Student at the University of Alberta)

- **Mohayeminul Islam, **Ajay Kumar Jha**, May Mahmoud, and Sarah Nadi. MigrateLib: A Tool for End-to-End Python Library Migration. *Empirical Software Engineering (EMSE)*, 2026.
- *David Owuor, **Ajay Kumar Jha**. Developer vs. DSpot vs. ChatGPT: A Comparative Study of JUnit Test Amplification. In the *19th IEEE International Conference on Software Testing, Verification and Validation (ICST)*, pp. 256-260, 2026. SVE Track.
- **Mohayeminul Islam, **Ajay Kumar Jha**, May Mahmoud, Ildar Akhmetov, and Sarah Nadi. An Empirical Study of Python Library Migration Using Large Language Models. In the *40th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, pp. 867-879, 2025.
- *Suraj Bhatta, *Frank Kendemah, **Ajay Kumar Jha**. Understanding Test Deletion in Java Applications. In the *22nd IEEE/ACM International Conference on Mining Software Repositories (MSR)*. pp. 408-420. 2025.
- **Ajay Kumar Jha**, Sarah Nadi. Migrating Unit Tests Across Java Applications. In the *24th IEEE International Conference on Source Code Analysis and Manipulation (SCAM)*. pp. 131-142. 2024.
- *Sai Kiran Bhrugumalla, **Ajay Kumar Jha**. TRec: A Regression Test Recommender for Java Projects. In the *40th IEEE International Conference on Software Maintenance and Evolution (ICSME)*. pp. 903-907. 2024. Tool Demo Track
- **Mohayeminul Islam, **Ajay Kumar Jha**, Ildar Akhmetov, and Sarah Nadi. Characterizing Python Library Migrations. In the *32nd ACM Symposium on the Foundations of Software Engineering (FSE)*. pp. 92-114. 2024.
- **Mohayeminul Islam, **Ajay Kumar Jha**, Sarah Nadi, and Ildar Akhmetov. PyMigBench: A Benchmark for Python Library Migration. In the *20th IEEE/ACM International Conference on Mining Software Repositories (MSR)*. pp. 511-515. 2023. Data and Tool Track.
- **Ajay Kumar Jha**, **Mohayeminul Islam, and Sarah Nadi. JTestMigBench and JTestMigTax: A benchmark and taxonomy for unit test migration. In the *30th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. pp. 713-717. 2023. ERA Track.
- Mansur Gulami, **Ajay Kumar Jha**, Sarah Nadi, Karim Ali, Emily Jiang, and Yee-Kang Chang. A Human-in-the-loop Approach to Generate Annotation Usage Rules: A Case Study with MicroProfile. In the *Annual International Conference on Computer Science and Software Engineering (CASCON)*, pp 91-100. 2022.
- **Batyr Nuryyev, **Ajay Kumar Jha**, Sarah Nadi, Yee-Kang Chang, Emily Jiang, and Vijay Sundaresan. Mining Annotation Usage Rules: A Case Study with MicroProfile. In the *38th IEEE International Conference on Software Maintenance and Evolution (ICSME)*. pp. 553-562. 2022. Industry Track
- **Ajay Kumar Jha** and Sarah Nadi. Annotation practices in Android apps. In the *20th IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM)*, pp. 132-142. 2020.
- Sooyong Jeong, **Ajay Kumar Jha**, Youngsul Shin, and Woo Jin Lee. A log-based testing approach for detecting faults caused by incorrect assumptions about the environment. *IEICE Transactions on Information and Systems*, 103(1), pp. 170-173. 2020
- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. An empirical study of configuration changes and adoption in Android apps. *Journal of Systems and Software (JSS)*. 156 (2019), pp. 164-180.
- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. Characterizing Android-specific crash bugs. In the *6th IEEE/ACM International Conference on Mobile Software Engineering and Systems (MobileSoft)*, pp. 111-122. 2019.
- **Ajay Kumar Jha**, Deok Yeop Kim, and Woo Jin Lee. A framework for testing Android apps by reusing test cases. In the *6th IEEE/ACM International Conference on Mobile Software Engineering and Systems (MobileSoft)*, pp. 20-24. 2019. Vision Track.
- **Ajay Kumar Jha** and Woo Jin Lee. An empirical study of collaborative model and its security risk in Android. *Journal of Systems and Software (JSS)*. 137 (2018), pp. 550-562.

- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. Developer mistakes in writing Android manifests: An empirical study of configuration errors. In the *14th IEEE/ACM International Conference on Mining Software Repositories (MSR)*, pp. 25-36. 2017.
- **Ajay Kumar Jha** and Woo Jin Lee. Analysis of permission-based security in android through policy expert, developer, and end user perspectives. *Journal of Universal Computer Science (JUCS)*, vol. 22, no. 4, pp. 459-474, 2016.
- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. Modeling and test case generation of inter-component communication in android. In the *Second ACM International Conference on Mobile Software Engineering and Systems (MobileSoft)*, pp. 113-116. 2015.
- **Ajay Kumar Jha**, Seungmin Lee, and Woo Jin Lee. Permission-based security in android application: from policy expert to end user. In the *2015 Research in Adaptive and Convergent Systems (RACS)*, pp. 319-320. ACM, 2015.
- Soo Young Jang, **Ajay Kumar Jha**, and Woo Jin Lee. Virtual prototype generation by shockwave flash for simulating HW components of embedded system. In the *29th Annual ACM Symposium on Applied Computing (SAC)*, pp. 1755-1756. 2014.
- **Ajay Kumar Jha** and Woo Jin Lee. Activity-based Event Capture and Replay Technique for Reproducing Crashes in Android Applications. *IEMEK Journal of Embedded Systems and Applications*. 9 (1), 1-9. 2014
- **Ajay Kumar Jha**, Sooyong Jeong, and Woo Jin Lee. Value-deterministic search-based replay for android multithreaded applications. In the *2013 Research in Adaptive and Convergent Systems (RACS)*, pp. 381-386. ACM, 2013.
- **Ajay Kumar Jha** and Woo Jin Lee. Capture and replay technique for reproducing crash in android applications. In the *12th IASTED International Conference in Software Engineering*, pp. 783-790. 2013.

Student Supervision

Current Students

Ph.D. Students

- **Kunal Dhanaikar**, North Dakota State University (since Sep 2026)
- **Frank Kendermah**, North Dakota State University (since Jan 2026)

Master's Students

- **Chloe Chinault**, North Dakota State University (since Sep 2025)
- **Aayush Lamsal**, North Dakota State University (since Feb 2025)

Alumni (*Co-advised with Dr. Sarah Nadi)

Ph.D. Students

- ***Mohayeminul Islam**, University of Alberta (graduated in Aug 2025)
- **Rupinder Kaur**, North Dakota State University (Mar 2024 – Jun 2025)
- **Amjad Allobadi**, North Dakota State University (Aug 2022 – Dec 2023)

Master's Students

- **Suraj Pokhrel**, North Dakota State University (graduated in Dec 2025)
- **David Owuor**, North Dakota State University (graduated in Dec 2025)
- **Frank Kendermah**, North Dakota State University (graduated in Dec 2025)
- **Suraj Bhatta**, North Dakota State University (graduated in Dec 2024)
- **Sai Kiran Bhrugumalla**, North Dakota State University (graduated in Jul 2024)
- ***Batyr Nuryyev**, University of Alberta (graduated in Nov 2021)

Undergraduate Research Interns

- **Griffin Winstead**, NDSU (Jan 2026 – May 2026), Supported by NDSU Explore Program (PI, \$2,781)
- **Xichen Pan**, University of Alberta (May 2021 – Sep 2021)

Teaching Experience (Course materials are available: <https://hifromajay.github.io/teaching.html>)

NDSU (*Newly developed course, **Substantially revised course)

- ***CSCI 412:** Mobile Software Engineering, Fall 2023, 2024, 2025, 2026
- ***CSCI 712:** Mobile Software Engineering, Spring 2025, 2026
- ****CSCI 713:** Software Development Processes, Fall 2025, 2026
- ****CSCI 714:** Software Project Planning and Estimation, Fall 2022, 2023, 2024
- ***CSCI 783:** Topics in Software Systems, Spring 2023, 2024
- ***CSCI 790:** Graduate Seminar:
 - Optimizing Regression Test Suites – Spring 2024
 - LLMs for Software Testing and Maintenance – Fall 2024, Spring 2026
 - Code Smell and Refactoring – Spring 2025

Professional Service

Conference Session Chair

- ICSME:** International Conference on Software Maintenance and Evolution - 2024
- SANER:** International Conference on Software Analysis, Evolution and Reengineering – 2022
- MSR:** International Conference on Mining Software Repositories - 2022

Program Committee Member

- MSR:** International Conference on Mining Software Repositories (Technical track) – 2021, 2022, 2024, 2025, 2026, 2027
- ICSME:** International Conference on Software Maintenance and Evolution (Technical track) - 2024, 2025, 2026
- SANER:** International Conference on Software Analysis, Evolution and Reengineering (ERA & Demo tracks) – 2021, 2022, 2024
- EASE:** International Conference on Evaluation and Assessment in Software Engineering (Research track) - 2026
- MobileSoft:** International Conference on Mobile Software Engineering and Systems (Research track) - 2022, 2023
- FSE:** International Symposium on the Foundations of Software Engineering (Artifact track) – 2021, 2022
- ASE:** International Conference on Automated Software Engineering (Artifact track) – 2021, 2022
- ICSE:** International Conference on Software Engineering (Demo track) – 2023

Journal Reviewer

- TSE:** IEEE Transactions on Software Engineering – 2020, 2021, 2022, 2024, 2025
- TOSEM:** ACM Transactions on Software Engineering and Methodology – 2025, 2026
- ESE:** Empirical Software Engineering - 2021, 2022
- SPE:** Journal of Software: Practice and Experience – 2020

Institutional Service (NDSU)

- CS Scholarship Committee,** Member (2023 – Present)
- College of Engineering Research & Graduate Committee,** Member (2025 – Present)
- College of Engineering Library Committee,** Representative (2024 – Present)
- Institutional Review Board (IRB),** Alternate Member (2023 – Present)

Invited Talks

Improving Software Quality: Strategies, Techniques, and Tools for Test Evolution, IEEE Siouxland Section Computer Society Speaker Event, South Dakota State University, Dec. 2025

Awards and Honors

National Research Foundation of Korea Postdoctoral Fellowship (March 2017 – February 2020)
Postdoctoral Researcher Funding supported by the National Research Foundation of Korea

Brain Korea 21 Plus Fellowship (September 2011 – February 2017)

Graduate research assistantship support during M.S. and Ph.D. studies

KNU Honors Scholarship (September 2013 – August 2015)

Merit-based scholarship awarded by Kyungpook National University for Ph.D. study

KNU Honors Scholarship (September 2011 – February 2013)

Merit-based scholarship awarded by Kyungpook National University for M.S. study